

Mesh-to-Mesh Transfer Errors in BDF2 Unsteady Flow Simulations

Abstract

Discrete remeshing in a time-accurate CFD calculation changes both the spatial discretization and the histories supplied to a multistep integrator. This work separates the resulting errors through static transfer tests, repeated Euler remeshing, controlled BDF2 restarts, and fixed-body RANS calculations with SA and SST turbulence transport. Nearest-neighbor Taylor transfer, barycentric interpolation, and continuous Galerkin projection recover their nominal first- to third-order smooth-field behavior. Under repeated Riemann remeshing, the dominant stress-test error is cumulative profile broadening rather than feature translation: after 400 transfers, NN1 and BAR2 broaden the density transition by 21.4 and 3.24 cells at the baseline deformation and by 26.4 and 22.9 cells when the displacement bound is doubled, whereas GP2 changes the width by at most 0.06 cell; the contrast is reproduced by three independent deformation realizations. A bounded global correction reduces the NN1 density-integral defect to roundoff without removing this broadening. In a controlled Euler NACA 0012 restart, perturbing only the older BDF2 history produces a finite load response even with an exact latest history, and the measured directional response ratios recover the BDF2 coefficients to within a few tenths of a percent in norm. The residual audit further shows that slip-wall enforcement modifies the wall-normal momentum component introduced by Galerkin transfer. In RANS, GP2 reduces the monitored turbulence-inventory defects by factors of roughly 37–247 relative to NN2, but the paired load difference between the two transfers, about 10^{-5} in lift, is insensitive to a doubling of the dual-time iteration cap and changes by more than its own magnitude when the physical step is halved, so no aerodynamic ranking is resolved. The results show that interpolation order, boundedness, integral preservation, multistep-history consistency, and receiving-solver convergence are distinct requirements; none alone predicts the aerodynamic consequence of remeshing.

1 Introduction

Unsteady Reynolds-averaged Navier–Stokes (RANS) calculations use mesh adaptation to follow evolving flow structures without maintaining fine resolution throughout the domain. When adaptation regenerates connectivity, the flow and turbulence histories must be transferred to the new discretization before time integration can continue. The resulting calculation combines spatial approximation, solution transfer, and temporal integration; improving one component does not necessarily reduce their combined error. Time-accurate anisotropic adaptation and space–time adaptation for implicit RANS calculations illustrate this coupling [1, 2].

Two questions arise at each mesh change. Does the transfer represent the source field accurately on the target mesh, and does the transferred state permit a consistent continuation of the target discrete evolution? Polynomial reproduction and integral conservation address the first question only partially. A transfer can preserve a total integral while redistributing the field in a way that changes a residual balance or an aerodynamic load. Conversely, an accurate pointwise interpolant can introduce a small global imbalance. Repeated transfers may accumulate either effect, depending on the error’s spatial structure and its interaction with the evolving flow.

The distinction between interpolation accuracy and preservation of a discrete state is well established. Common-refinement Galerkin projection provides an integral-preserving, best- L_2 transfer between finite-element spaces [3, 4], while constrained variants are needed when additional discrete properties must be preserved [5]. Balance-preserving transfer in shallow-water models provides a related example: an accurate interpolant need not preserve a steady discrete balance [6, 7]. Here these ideas are examined in the specific setting of a vertex-centered finite-volume BDF2 restart and its subsequent aerodynamic response.

The comparison uses three established constructions: nearest-neighbor Taylor transfer, barycentric interpolation on a simplicial representation of the source mesh, and continuous Galerkin projection. Their nominal orders range from one to three. A bounded global correction is also applied to the pointwise methods, allowing the effect of enforcing a total integral to be assessed without replacing the underlying interpolation rule. The analysis distinguishes the consistent finite-element integral from the nodal integral actually used by the receiving solver, particularly when a quadratic projected field is reduced to vertex-only output.

The numerical argument proceeds from interpolation verification to repeated transport and then to the response of the receiving flow solver. A two-dimensional Euler Riemann problem isolates accumulation under changes in remeshing frequency and deformation amplitude. A controlled Euler NACA 0012 restart separates the contributions of the two BDF2 histories and examines boundary-condition processing and delayed aerodynamic response. The final application retains a fixed airfoil but adds SA and SST turbulence transport, viscous stresses, and repeated random mesh replacement. This sequence is designed to determine which transfer properties survive contact with a complete time-accurate CFD pipeline, rather than to establish a universal ranking of interpolation operators. Continuous mesh motion and moving bodies are outside the present scope.

2 Solution-transfer operators

2.1 Representation and reconstructed derivatives

The source and target meshes, A and B , cover the same discretized domain Ω . Their nodal values are U_A and U_B , and $M_M = \text{diag}(w_i^M) \otimes I$ contains the positive control-volume weights of the vertex-centered flow solver on mesh M . For one component ϕ , the solver’s integral is

$$Q_M(\phi_M) = \sum_i w_i^M \phi_{M,i}. \quad (1)$$

All compared transfers return values at the solver vertices. Density, momentum, and total energy are transferred in conserved-variable form; nonlinear products are not reconstructed from separately interpolated primitive variables. Nodal sampling and control-volume averaging are different representation operators, even when the same weights are used to monitor an integral.

Nearest-neighbor corrections and the quadratic source representation require derivatives recovered from surrounding nodal data. A weighted least-squares fit determines the coefficients in

$$\phi_j - \phi_i \simeq g_i \cdot r_{ij} + \frac{1}{2} r_{ij}^T H_i r_{ij}, \quad r_{ij} = x_j - x_i. \quad (2)$$

The degree-one fit omits the Hessian; the degree-two fit retains it. With a full-rank, suitably scaled stencil and smooth data, polynomial reproduction gives an $O(h)$ gradient error for the linear fit and $O(h^2)$ gradient and $O(h)$ Hessian errors for the quadratic fit. These are conditional estimates: irregularity, anisotropy, rank deficiency, and weighting can degrade derivative recovery [8, 9, 10, 11]. The third-order variants therefore use degree-two reconstruction. Their accuracy is checked numerically rather than inferred from the requested polynomial degree alone.

2.2 Nearest-neighbor and barycentric interpolation

Nearest-neighbor transfer selects the closest source vertex $n(j)$ to each target vertex and evaluates a Taylor expansion there. With $r_j = x_j^B - x_{n(j)}^A$, the three variants are

$$(\mathcal{I}_{AB}^{\text{NN1}} \phi_A)_j = \phi_{A,n(j)}, \quad (3)$$

$$(\mathcal{I}_{AB}^{\text{NN2}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j, \quad (4)$$

$$(\mathcal{I}_{AB}^{\text{NN3}} \phi_A)_j = \phi_{A,n(j)} + g_{n(j)} \cdot r_j + \frac{1}{2} r_j^T H_{n(j)} r_j. \quad (5)$$

Thus NN1 copies a value, NN2 accounts for local slope, and NN3 also accounts for curvature. No containing-element search is required. For $|r_j| = O(h)$ and accurate derivatives, their smooth-field errors are respectively $O(h)$, $O(h^2)$, and $O(h^3)$. The selected donor changes discontinuously across source-node Voronoi boundaries, so a smooth displacement need not produce a smooth change in the interpolation error.

Barycentric interpolation instead locates the source simplex containing the target point. A conforming triangulated or tetrahedralized view of the original mesh supplies this search geometry, using original vertices rather than an augmented control-volume point set. If the containing simplex has vertices $x_{i_r}^A$, its barycentric coordinates satisfy

$$x_j^B = \sum_{r=1}^{d+1} \lambda_r x_{i_r}^A, \quad \sum_r \lambda_r = 1, \quad \lambda_r \geq 0. \quad (6)$$

They express the target point as a geometric weighted average of the simplex vertices. Applying the same weights to the field gives

$$(\mathcal{I}_{AB}^{\text{BAR2}} \phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \quad (7)$$

$$(\mathcal{I}_{AB}^{\text{BAR3}} \phi_A)_j = \sum_r \lambda_r \left[\phi_{A,i_r} + \frac{1}{2} g_{i_r} \cdot (x_j^B - x_{i_r}^A) \right]. \quad (8)$$

BAR2 is affine-exact and already second order. For a quadratic field with exact vertex gradients, the half-gradient term in BAR3 cancels the quadratic interpolation remainder, yielding third-order consistency. The factor $1/2$ follows from this cancellation rather than from calibration. Both expressions are continuous across a conforming simplex interface when shared vertices have unique values and gradients. Neither enforces Eq. (1) between meshes.

Limiting is separate from nominal order. A factor $0 \leq \Phi_i \leq 1$ may scale both derivative terms in NN3, so $\Phi_i = 0$ recovers NN1; in BAR3 it scales the gradient correction and recovers BAR2, not a piecewise-constant interpolant. The smooth verification uses unlimited reconstruction. The discontinuous tests compare unlimited and Barth–Jespersen-limited transfer [12]; differentiable and higher-order limiter constructions provide related alternatives [13, 14, 15]. Damping a quadratic with a factor chosen from linear increments alone does not establish boundedness at every evaluation point. The measured extrema and the final thermodynamic admissibility of the state remain necessary diagnostics.

2.3 Galerkin projection and common-refinement integration

Galerkin projection determines a coupled target field rather than evaluating each target point independently. Let V_B^p be the continuous degree- p Lagrange space on the target simplicial mesh. The projection of a source representation ϕ_A^h satisfies

$$(\phi_B^h - \phi_A^h, v_B)_{L_2(\Omega)} = 0 \quad \forall v_B \in V_B^p, \quad (9)$$

and minimizes the L_2 distance to that source representation [3, 4]. For $\phi_B^h = \sum_j v_j N_j^B$, this gives

$$H_B v = b, \quad (H_B)_{ij} = \int_{\Omega} N_i^B N_j^B dx, \quad b_i = \int_{\Omega} N_i^B \phi_A^h dx. \quad (10)$$

The consistent mass matrix H_B is sparse and positive definite. Its off-diagonal entries couple basis functions sharing target elements; solving the system determines all coefficients simultaneously. It is distinct from the diagonal finite-volume matrix M_B used in time integration.

The load vector is the cross-mesh operation. On one target element, N_i^B has a fixed polynomial expression, but the source field may cross several source-element boundaries and change polynomial expression within that element. Consequently,

$$b_i = \sum_{K_B} \sum_{K_A: |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N_i^B \phi_A^h dx. \quad (11)$$

Each intersection fixes both polynomial expressions and permits exact polynomial quadrature. These intersections form the source–target common refinement; they may be integrated locally without storing a complete additional mesh. Coverage must be complete and nonoverlapping. Products require degree-two quadrature for P^1/P^1 transfer and degree-four quadrature for P^2/P^2 transfer on affine simplices [16, 17]. This geometric integration and the mass-matrix solve distinguish projection from the point searches in NN and BAR.

GP2 uses P^1 source and target spaces. GP3 uses P^2 spaces, introducing edge coefficients internally. With $d_{ij} = x_i - x_j$ and $\bar{H}_{ij} = (H_i + H_j)/2$, the source midpoint value is reconstructed as

$$\phi_{ij} = \frac{1}{2}(\phi_i + \phi_j) - \frac{1}{8}d_{ij}^T \bar{H}_{ij} d_{ij}. \quad (12)$$

For a quadratic, the subtracted curvature term is exactly the difference between the endpoint average and the midpoint value. GP3 then solves for the complete quadratic target field, but only its original-vertex values are supplied to the flow solver. Nominal L_2 approximation orders are two and three under the usual smoothness and mesh-stability assumptions; nodal accuracy after this complete pipeline is assessed separately.

On mixed-element grids, BAR and GP use conforming subdivisions based on a deterministic shared-face diagonal rule. The approximation is P^p on the simplices, not Q^p on the native quadrilaterals or hexahedra. A valid conforming split and the chosen control-volume weights must therefore be retained consistently in geometric and integral checks.

2.4 Which integral is conserved?

For an unconstrained target space containing constants, Eq. (9) preserves the integral of the complete finite-element field. Coordinate functions also belong to continuous P^1 , and quadratic coordinates belong to P^2 . Exact projection therefore preserves the corresponding consistent moments before constraints, sampling, or limiting. This is not finite-volume conservation over arbitrary target cells: their indicator functions generally do not belong to the continuous test space. Nor is the inverse mass matrix a local transfer operator. Equality of source and target geometry on one subregion does not by itself make the projection the identity there.

For GP2, the consistent and solver-lumped integrals coincide when $w_i^M = \int N_i^M dx$ on both meshes. For GP3, source lifting and target vertex restriction each introduce a representation-dependent difference. In particular, a third-order projected field does not imply an $O(h^3)$ lumped-integral defect. A simple quadratic counterexample is given in Appendix A.1: exact P^2 projection can still produce an $O(h^2)$ error after restriction to vertex values and P^1 lumped integration. The

consistent projection, the stored restart field, and any repaired field are consequently distinguished throughout the comparison.

A global correction can enforce a chosen solver integral independently of the transfer construction. For a raw nodal field ϕ_B^r , it seeks a correction c such that

$$\sum_j w_j^B c_j = Q_A(\phi_A) - Q_B(\phi_B^r), \quad \ell_j \leq \phi_{B,j}^r + c_j \leq u_j. \quad (13)$$

The implemented repair redistributes the defect over nodes with available capacity, freezing saturated nodes [18, 19]. Feasibility requires the requested integral to lie between the weighted lower and upper bounds, with fixed boundary values included. Source-range bounds on individual conserved components do not alone guarantee positive pressure. Accuracy is retained only if the correction is sufficiently small in the accuracy norm; a small total defect does not itself bound every nodal increment.

The optional NN/BAR correction is denoted by the suffix GC. GP additionally uses a projection-level limiter: source-vertex values in overlapping elements define $[\ell_j, u_j]$, widened by τ times the global source range. Clipping is followed by redistribution preserving a selected lumped total. Hard limiting sets $\tau = 0$; soft limiting allows a controlled excursion. Once active, this nonlinear post-processing need not preserve Galerkin orthogonality, first moments, or the best-approximation property. For GP2, preserving its lumped total preserves its consistent mass only under the weight compatibility above. For vertex-only GP3, no such implication holds automatically. These qualifications matter when interpreting the method labels in the numerical results.

3 BDF2 response to a mesh change

3.1 Reference trajectories and history mismatch

Between remeshing events, the mesh is fixed and the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \quad (14)$$

The residual can contain Euler, viscous, or RANS terms; the state must include every transported variable required by the time integrator. The constant-step BDF2 equation on the target mesh is

$$\Phi_B(Y; V, W) = \frac{M_B}{\Delta t} \left(\frac{3}{2}Y - 2V + \frac{1}{2}W \right) + F_B(Y, t^{n+1}) = 0. \quad (15)$$

This form is not a general moving-volume ALE formula. Continuously moving meshes require consistent geometric history terms and a discrete geometric conservation law [20, 21]; those are outside the fixed-mesh-between-transfers derivation.

Let U_A^k and U_B^k be native trajectories for the same physical problem, computed on the two meshes. The transferred histories and their target-referenced discrepancies are

$$\hat{U}_B^k = \mathcal{I}_{AB}(U_A^k), \quad \varepsilon^k = \hat{U}_B^k - U_B^k, \quad k = n, n-1. \quad (16)$$

Here $\mathcal{I}_{AB}$ includes all limiting, correction, and restart-field processing. Let $\Pi_M u^k$ represent a common exact field in the solver's chosen data representation, and $e_M^k = U_M^k - \Pi_M u^k$. An exact decomposition, valid also for nonlinear transfer, is

$$\begin{aligned} \varepsilon^k = & \underbrace{\mathcal{I}_{AB}(\Pi_A u^k) - \Pi_B u^k}_{\text{transfer consistency}} \\ & + \underbrace{\mathcal{I}_{AB}(U_A^k) - \mathcal{I}_{AB}(\Pi_A u^k) - e_B^k}_{\text{propagated source error and target error}}. \end{aligned} \quad (17)$$

Only for a linear transfer may the second line be replaced by $\mathcal{I}_{AB}e_A^k - e_B^k$.

Thus a native-target reference does not make ε^k a pure interpolation error. It puts both states into the same target evolution problem while retaining their pre-existing discretization mismatch. For an output G , the decomposition

$$\begin{aligned} G_B(\hat{U}_B^k) - G_A(U_A^k) &= [G_B(U_B^k) - G_A(U_A^k)] \\ &\quad + [G_B(\hat{U}_B^k) - G_B(U_B^k)] \end{aligned} \quad (18)$$

separates native mesh sensitivity from the target-referenced restart discrepancy, not from an independently known consistency error. Comparisons of different transfers from the same source histories to the same target provide the controlled method comparison.

3.2 Exact history forcing and local state response

At any common trial state Y , changing only the histories gives the exact identity

$$\Phi_B(Y; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(Y; U_B^n, U_B^{n-1}) = -\frac{M_B}{\Delta t} \eta^n, \quad \eta^n = 2\varepsilon^n - \frac{1}{2}\varepsilon^{n-1}. \quad (19)$$

This identity does not require a linear spatial residual or small transfer errors. It tests history assembly provided both residual evaluations use exactly the same trial state and residual convention.

For the solved states, define $\delta^{n+1} = \hat{U}_B^{n+1} - U_B^{n+1}$. Let r and $\hat{r}$ be the remaining BDF2 algebraic residuals. Subtraction and expansion about the native target state give

$$A_B \delta^{n+1} = M_B \eta^n + \Delta t(\hat{r} - r) - \Delta t \mathcal{N}_B(\delta^{n+1}), \quad A_B = \frac{3}{2}M_B + \Delta t J_B, \quad (20)$$

where $J_B = \partial F_B / \partial U$ and $\mathcal{N}_B$ is the nonlinear Taylor remainder. With a locally Lipschitz Jacobian, $\mathcal{N}_B(\delta) = O(\|\delta\|^2)$. If A_B is locally invertible, the perturbations are small, and algebraic errors are negligible,

$$\delta^{n+1} = R_B \eta^n + O(\Delta t \|\eta^n\|^2), \quad R_B = A_B^{-1} M_B. \quad (21)$$

The constants are local to the mesh, state, time step, and norm; they are not assumed uniform over arbitrary anisotropic mesh families. Active limiter switches or insufficient inner convergence may invalidate this smooth-response approximation, but not Eq. (19).

The steady experiment uses one spatial discrepancy $\varepsilon^* = \mathcal{I}_{AB}U_A^* - U_B^*$ and assigns it independently to the two histories:

History configuration	ε^n	ε^{n-1}	η^n
Native	0	0	0
Full	ε^*	ε^*	$3\varepsilon^*/2$
Latest only	ε^*	0	$2\varepsilon^*$
Older only	0	ε^*	$-\varepsilon^*/2$

These diagnostic restarts predict

$$\delta_n \simeq \frac{4}{3}\delta_{\text{full}}, \quad \delta_{n-1} \simeq -\frac{1}{3}\delta_{\text{full}}, \quad \delta_{\text{full}} \simeq \delta_n + \delta_{n-1}. \quad (22)$$

The older-only case distinguishes the two-history forcing from a latest-state diagnostic. All three perturbations nevertheless use the same spatial direction: recovering their ratios is a test of directional response, not a complete validation of R_B for arbitrary history fields.

3.3 Aerodynamic response and propagation

For a differentiable aerodynamic functional with gradient g_B^{n+1} , the first-step discrepancy is

$$\Delta G^{n+1} = (g_B^{n+1})^T R_B \eta^n + O(\|\eta^n\|^2). \quad (23)$$

Equivalently, $A_B^T z_B = g_B^{n+1}$ gives $\Delta G^{n+1} = z_B^T M_B \eta^n$ to first order [22]. The error is weighted by the target dynamics and the output sensitivity, not by a constant conservation test. Conservation of the source integral constrains only

$$\mathbf{1}^T M_B \varepsilon^k = Q_A(U_A^k) - Q_B(U_B^k), \quad (24)$$

which need not vanish because the native integrals may differ.

A slope discrepancy must include the output difference already present at the transfer time:

$$\frac{\Delta G^{n+1} - \Delta G^n}{\Delta t} \simeq \frac{(g_B^{n+1})^T R_B \eta^n - (g_B^n)^T \varepsilon^n}{\Delta t}. \quad (25)$$

Dividing the first-step output difference alone by Δt is generally not this quantity. Cancellation between the two terms also prevents a universal force-kink scaling from being inferred from interpolation order alone.

For later steps, linearization along the native target trajectory gives

$$\begin{bmatrix} \delta^{k+1} \\ \delta^k \end{bmatrix} \simeq \begin{bmatrix} 2R_B^{k+1} & -R_B^{k+1}/2 \\ I & 0 \end{bmatrix} \begin{bmatrix} \delta^k \\ \delta^{k-1} \end{bmatrix}. \quad (26)$$

Repeated application connects the initial histories to a delayed output. This explains why a first-step force difference need not bound the later excursion. A dominant disturbance originating away from the airfoil may become measurable after acoustic or convective transport, but neither the inverse mass matrix of projection nor an implicit time step has compact support. Distance-dependent arrival is therefore measured by amplitude thresholds, not by a claim of strict first-nonzero causality.

4 Numerical experiments and assessment

4.1 Static transfer and repeated Euler transport

The static tests separate transfer error from flow-solver error. The asymmetric smooth density is

$$\phi = \rho_\infty \left[2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y) \right]. \quad (27)$$

The field is transferred to structured quadrilateral and independently generated mixed triangular/quadrilateral target meshes at nominal resolutions 64, 128, and 256; the source mesh has half the target nominal resolution. Errors are relative nodal density L_2 differences from the analytic value at the target points. The effective spacing is defined as $h_{\text{eff}} = N_{\text{nodes}}^{-1/2}$; on this fixed domain it is used only to define refinement ratios. The discontinuous test replaces the smooth field with a circular inclusion of radius $0.25 \min(L_x, L_y)$ and density levels 1.2 and 3.6. Because the curved interface is not mesh-aligned, unlimited and limited transfers can be compared on a common unresolved discontinuity.

Consecutive and fitted orders are obtained from

$$p_{k,k+1} = \frac{\log(E_k/E_{k+1})}{\log(h_k/h_{k+1})}, \quad \log E \simeq p_{\text{fit}} \log h + b. \quad (28)$$

The three-level sequences are used to check consistency with the nominal orders, not to estimate mesh-independent asymptotic constants. A second smooth test on the structured Riemann mesh family keeps the mean displacement-to-spacing ratio near 0.057 and is used only as a consistency check on the implemented operators.

The unsteady Riemann calculation solves two-dimensional Euler flow on $[0, 5] \times [0, 1]$, with a uniform 501×101 -node quadrilateral mesh, $h = 0.01$, $\Delta t = 10^{-3}$, and 400 physical steps. At each event, the original mesh receives a new bounded random displacement in x , with boundary vertices fixed. Distortion is not accumulated geometrically from one event to the next. The bound is the minimum incident corner-triangle altitude divided by SafeFactor, denoted SF. Two transfers per block give 80 transfers for a ten-step interval or 400 for a two-step interval. SF4 is the baseline; SF2 doubles the displacement bound. A no-remeshing calculation supplies a fixed-mesh reference. This experiment measures the combined effect of changing spatial discretization and transferring the solution, not a pure interpolation error relative to an exact flow.

Interpolation-only integral changes are sampled immediately before and after each transfer and accumulated separately from the physical evolution. Density and energy are used because their reference integrals are away from zero. The recomputed stress-test dataset uses one common y -averaged profile extractor for every frequency and amplitude comparison: the transition position is obtained from the 50% crossing and its width from the 10–90% interval, both reported relative to the no-remeshing reference in units of the baseline cell size. This common extraction replaces the earlier mixed-tracker comparison. The transition remains a broad density feature rather than an independently validated sharp shock. The frequency and correction matrix uses one deformation sequence; the NN1, BAR2, and GP2 entries at 400 transfers are repeated with two further random realizations at both amplitudes.

4.2 NACA 0012: independent meshes and controlled histories

The aerodynamic case is Euler flow over NACA 0012 at $M_\infty = 0.1$ and $\alpha = 6^\circ$, with $p_\infty = \rho_\infty = T_\infty = 1$, $a_\infty = \sqrt{1.4} = 1.18322$, and $U_\infty = 0.118322$. BDF2 dual-time integration uses $\Delta t = 10^{-3}$. Forces are pressure-only, with unit reference chord and area and moment reference $x/c = 0.25$. The nonzero steady computed drag is treated as a numerical load, not as skin friction or shock-induced wave drag; absolute coefficient changes are used.

The independent source and target meshes contain 16,255/32,077 and 22,508/44,583 nodes/triangles, respectively. Both share the same 370 airfoil vertices and 63-node farfield polygon at 100 chords. The surface geometry is therefore unchanged although the interior meshes differ. Each native steady solution is advanced with equal BDF2 histories to create its own reference trajectory. Transfers from A to B then populate the history configurations of Eq. (22). All comparisons of their solved responses use the native target trajectory at the same physical step.

State discrepancies use the nondimensional volume-weighted norm $\|v\|_M = (v^T M v)^{1/2}$, summed over the stored flow components. Because the farfield occupies most of the domain, this norm emphasizes the global state and is complemented by near-wall and force diagnostics. Vector superposition is assessed by

$$e_{\text{sup}} = \frac{\|\delta_{\text{full}} - \delta_n - \delta_{n-1}\|_{M_B}}{\|\delta_{\text{full}}\|_{M_B}}, \quad (29)$$

with the analogous scalar expression for lift. Signed force ratios and positive state-norm ratios are kept distinct.

The solver exports $-\Phi_B$, including the temporal residual. Initial residuals are compared before the first update and only for cases sharing a trial state: older-only versus native at U_B^* , and latest-only versus full at $\hat{U}_B^*$. Their $\Delta\eta$ values are respectively $-\epsilon^*/2$ and $+\epsilon^*/2$. A dedicated double-precision

residual probe is used for the final identity check. Because the NACA boundary is a slip wall, the residual is also decomposed into wall-normal and wall-tangential momentum components to distinguish the raw transferred history from the boundary-admissible state. First-step forces and subsequent 50-step summaries provide separate assessments of immediate and later response.

4.3 Repeated airfoil remeshing and localized disturbances

The repeated-remeshing experiment perturbs the source mesh between wall distances 0.01 and 20 chords while keeping the boundaries fixed. Thirty-nine mesh replacements occur, one every ten physical steps in the 400-step window. NN2 and GP2 are evaluated on the same two random deformation sequences, and force differences are formed relative to the no-remeshing trajectory. Least-squares slopes are quoted per 100 physical steps as descriptive summaries; the associated regression standard errors refer to each individual correlated time series and are not ensemble uncertainties.

A separate one-event experiment perturbs narrow wall-distance bands centered at $d/c = 0.05$, 0.10, and 0.20, then continues without further adaptation. Force onset is the first crossing of a selected fraction of that case’s peak response. Fixed pressure shells provide a less spatially integrated diagnostic through their root-mean-square pressure difference. A shell a distance $d - r$ inside the launch band has the approximate acoustic lag $(d - r)/(a_\infty \Delta t) \simeq 845(d - r)$. This is a low-Mach scale estimate; actual characteristics depend on local velocity, sound speed, and propagation direction.

The no-remeshing trajectory is the reference for this localized-event experiment, whereas the mixed-history study uses an independently evolved target-mesh reference. Even though the meshes coincide outside the displaced band, the transfer is not guaranteed to remain strictly local. An immediate response is observed up to about $0.08c$ inward of the band; those shells are excluded from the propagation fit. Projection coupling, limiter redistribution, the changed spatial operator, and the implicit update can all contribute to this near-instantaneous component.

4.4 Fixed-body RANS extension

The RANS extension uses the fixed NACA 0012 at $M_\infty = 0.1$, $\alpha = 6^\circ$, and $Re_c = 6 \times 10^6$ with Spalart–Allmaras (SA) and shear-stress-transport (SST) closures [23, 24]. The common hybrid mesh contains 23,271 vertices, 9,686 quadrilaterals in the wall layer and wake block, and 26,851 triangles, with 256 airfoil vertices and a farfield at $100c$. The first off-wall spacing is about $5 \times 10^{-6}c$ and the quadrilateral region extends to approximately $0.11c$. The experiment is not a turbulence-model validation; it asks how repeated mesh replacement perturbs a nominally steady viscous solution when the turbulence histories are transferred with the mean flow.

The executed SU2 configurations use HLLC mean-flow fluxes with MUSCL reconstruction, scalar upwinding without MUSCL for the turbulence equations, CFL 6, FGMRES/ILU, and a maximum of 800 dual-time iterations per physical step. The convergence test begins after ten inner iterations and targets $\log_{10}(\text{rms } \rho) = -12$. Runtime startup tables establish that the configured 10^{-3} s step corresponds to the non-dimensional value $\Delta t^* = 0.347224$ under `FREESTREAM_VEL_EQ_MACH`; with $U_\infty^* = 0.1$, the 400-step window spans 13.9 chord flow-through times. The first-order startup step is ten times smaller. Each closure has an 800-step no-remeshing reference initialized from the same restart histories.

At a repeated-remeshing event 12,602 vertices are movable and 10,669 remain fixed. No vertex with solver wall distance below $0.01c$ is displaced; the motion mask is based on distance to the nearest airfoil node, so some vertices in the outer part of the quadrilateral layer do move. Boundary vertices are fixed. SF4 defines the baseline displacement and a minimum corner-area ratio of 0.2

is enforced; the recomputed mesh-quality audit shows valid grids throughout the event sequence. The displacement field depends only on the prescribed seed and event index, not on the evolving solution. For the seed-1 comparison, archived displacement hashes verify identical mesh sequences across SA/SST and NN2/GP2 cases.

The repeated comparison uses NN2 and limited GP2 with two deformation realizations and a replacement every ten physical steps. Density, momentum, total energy, and both BDF2 histories of the turbulence variables are transferred. SA transports $\tilde{\nu}$; SST transfers density-weighted k and ω , after which the receiving solver applies its primitive-variable positivity floors. Wall distance and eddy viscosity are recomputed. The monitored transfer integrals are therefore properties of the returned restart pipeline at the recorded stage, not conservation laws for subsequent turbulence evolution.

Total aerodynamic coefficients are compared with the matching no-remeshing reference. Pressure and friction contributions are additionally integrated from the exported surface fields using identical quadrature for case and reference. The repeated-remeshing force histories contain 398 distinct physical steps. Because the physical-step solve is part of the measured response, the number of inner iterations and terminal residuals are analyzed together with the transfer data rather than assumed negligible.

4.5 Evidence and reproducibility limits

The interpolation-order data and the recomputed Riemann and Euler NACA summary tables are versioned in the repository. The RANS archive additionally contains force histories, surface reductions, transfer-integral records, mesh-quality diagnostics, runtime normalization, code provenance, and inner-iteration histories. These products permit the comparisons reported below to be regenerated, but they do not replace an independent rerun of every CFD case. The two airfoil realizations and the three Riemann realizations quantify reproducibility only at a pilot level. The RANS method comparison is interpreted together with three numerical controls on the same event sequence: a doubled dual-time iteration cap, a halved physical step with the events replayed at the same physical times, and a no-transfer baseline on a finer viscous mesh (Sec. 5.6).

5 Results

5.1 Interpolation order verification

Figure 1 shows the smooth-field refinement results. NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 have fitted slopes above three over the three-level interval, approaching approximately 3.2 on the finest pair. This is finite-range convergence behavior, not evidence of a higher designed order. On the finest mixed grid, the relative errors are approximately 2.6×10^{-7} for BAR3, 3.4×10^{-7} for GP3, and 2.6×10^{-6} for NN3. These differences resolve smooth interpolation performance on the tested mesh family; they do not predict a corresponding ranking of restarted flow solutions.

5.2 Boundedness and integral preservation are independent constraints

The discontinuous transfer test separates three properties that coincide only in special cases: approximation error, boundedness, and preservation of the solver integral. Table 1 reports representative configurations on the structured $64 \rightarrow 128$ transfer, the middle level of the refinement sequence. The second-order barycentric interpolant is bounded by construction on each containing simplex,

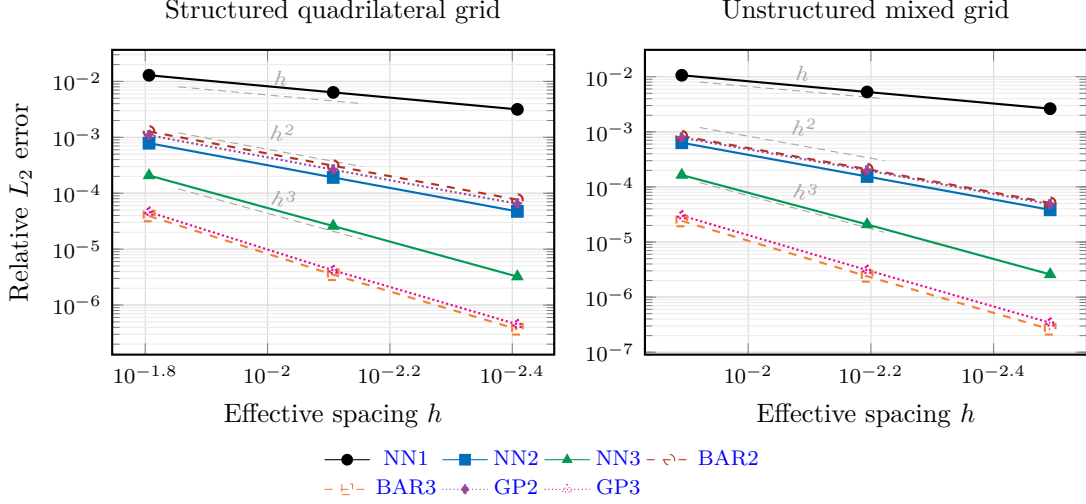


Figure 1: Convergence of the density-transfer error for the smooth asymmetric field. Curves show the relative nodal L_2 error as the mesh is refined; gray guides correspond to $O(h)$, $O(h^2)$, and $O(h^3)$.

Table 1: Representative discontinuous-field transfers. Overshoot and undershoot are excursions beyond the exact density range $[1.2, 3.6]$; the last column is the source-to-target defect in the solver’s lumped density integral. BJ denotes Barth–Jespersen limiting and GC the bounded global conservation correction.

Transfer	Overshoot	Undershoot	Integral defect
NN2, unlimited	0.8031	0.4157	-5.26×10^{-5}
NN2, BJ	0	0	-4.46×10^{-3}
NN2, BJ+GC	0	0	2.95×10^{-14}
BAR3, unlimited	0.1766	0.1811	-8.80×10^{-7}
BAR3, BJ+GC	0	0	5.59×10^{-14}
GP2, hard limited	0	0	3.44×10^{-15}
GP3, hard limited, vertex output	0	0	-4.83×10^{-7}

whereas the unlimited derivative-based reconstructions generate new extrema. For NN2 the largest excursion is 0.803 above the upper plateau and 0.416 below the lower plateau; BAR3 reduces these excursions to 0.177 and 0.181, but does not remove them. Barth–Jespersen limiting removes the measured extrema. The same limiting step does not make NN2 conservative: where the limiter is active the reconstruction collapses to NN1, and the density-integral defect grows to the NN1 value of -4.46×10^{-3} until the global correction is applied. Conversely, unlimited GP2 preserves its consistent integral to roundoff while remaining oscillatory. These observations rule out treating high order, boundedness, and conservation as interchangeable indicators of transfer quality.

The smooth single-transfer test reaches the same conclusion from the opposite direction. GC reduces the monitored density-integral defect to $O(10^{-15})$ for every base transfer while changing the density L_1 error by at most 0.8% at that resolution. It is therefore inexpensive in this smooth case, but its scalar constraint says nothing about where the correction is placed. The repeated-remeshing results below test that missing spatial information directly.

Table 2: Effect of remeshing frequency, deformation amplitude, and global correction on the recomputed density-transition diagnostic. Width changes and absolute position errors are expressed in baseline cells ($h = 0.01$). SF2 permits twice the displacement of SF4.

Condition	NN1 width	NN1+GC width	BAR2 width	GP2 width	max. $ \Delta x /h$
80 transfers, SF4	0.57	–	0.25	0.00	0.14
400 transfers, SF4	21.41	18.36	3.24	0.00	0.15
400 transfers, SF2	26.36	25.65	22.91	0.06	0.91

5.3 Repeated Euler remeshing: diffusion grows before position error

The recomputed Riemann diagnostics change the interpretation of the stress test. At the baseline 80-transfer setting, all methods locate the density transition within 0.14 cell of the no-remeshing reference. The largest width change is 0.57 cell for NN1 and 0.25 cell for BAR2; the higher-order NN/BAR variants and both Galerkin variants change the width by at most 0.01 cell. The dominant failure mode under more frequent or larger mesh changes is therefore not a systematic translation of the feature but cumulative broadening.

Table 2 makes this distinction explicit. Increasing the number of transfers from 80 to 400 at SF4 expands the NN1 transition from 0.57 to 21.41 cells beyond the reference width, while its position error remains 0.10 cell. BAR2 broadens by 3.24 cells, whereas GP2 remains unchanged to the reported precision. Doubling the admissible deformation at 400 transfers increases the width changes to 26.36 cells for NN1 and 22.91 cells for BAR2; GP2 changes by only 0.06 cell. Even in this most aggressive comparison the reported position errors remain below one cell for all entries in the table. Two further random deformation sequences reproduce these values (Table 6): across the three realizations the width change is 21.4–22.9 cells for NN1 and 3.1–3.2 cells for BAR2 at SF4, 25.0–27.1 and 22.9–23.0 cells at SF2, and at most 0.06 cell for GP2, with every position error below 0.5 cell.

The global correction provides the cleanest separation between integral and spatial error. For NN1 at 400 transfers, it reduces the cumulative density-integral change from 5.0×10^{-4} to 5.8×10^{-14} at SF4 and from 1.3×10^{-3} to 4.3×10^{-14} at SF2. The corresponding transition widths nevertheless remain 18.36 and 25.65 cells above the reference. Global conservation therefore removes the scalar integral defect without reconstructing the lost spatial distribution. This result does not imply that Galerkin projection is uniquely capable of preserving the profile: NN2, NN3, BAR3, and GP3 also remain sharply resolved in the 400-transfer SF4 comparison. It does show that a zero global integral defect is insufficient to diagnose repeated-remapping fidelity.

The same hierarchy is visible in the interpolation-only accumulation. At 80 transfers NN1 changes the density integral by +0.0105%, compared with +0.0020% for BAR2 and +0.0003% for GP2. At the more frequent SF4 setting the NN1 change reaches +0.0500%, while GP2 remains at −0.0003%. These signed totals can contain cancellation and are not used as drift rates; their role is to quantify the imbalance introduced at the transfer events themselves.

5.4 A BDF2 restart depends on both transferred histories

The steady Euler NACA 0012 experiment isolates the temporal consequence of a mesh transfer without introducing a physical unsteadiness. The target-referenced state discrepancy after transfer is approximately 7.0×10^{-4} for every second- and third-order method, with less than 3% spread, whereas NN1 gives 1.47×10^{-3} . This clustering is important: it prevents the subsequent response from being interpreted as a simple ranking of interpolation accuracy, because the target-referenced

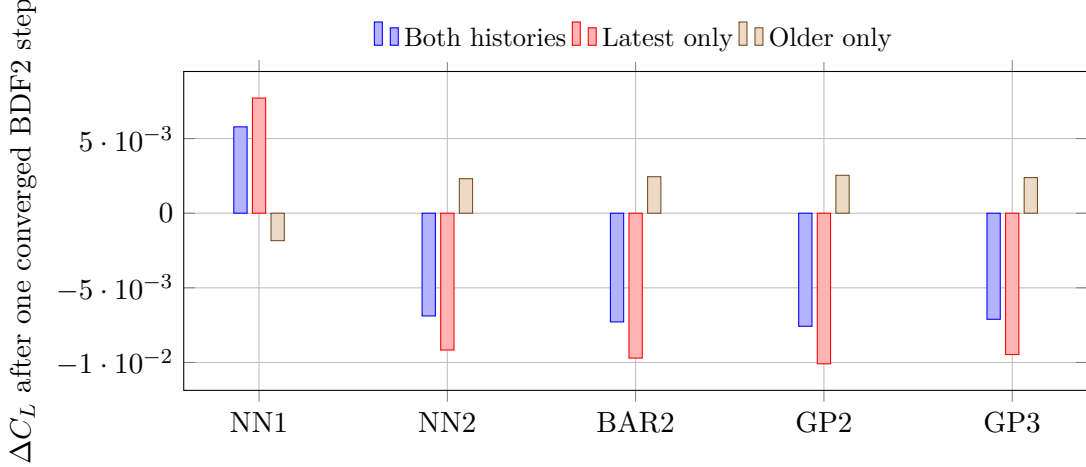


Figure 2: First-step lift response for the controlled Euler NACA 0012 BDF2 restart. The same target-referenced spatial discrepancy is inserted into both histories, only the latest history, or only the older history. The nonzero older-only response demonstrates that the latest state alone is not a sufficient restart diagnostic.

discrepancy also contains the native difference between the two discrete steady solutions.

The controlled history experiment nevertheless gives a direct result about BDF2. Figure 2 shows the first converged lift response when the same spatial discrepancy is placed in both histories, only the latest history, or only the older history. The older-only configuration has $\epsilon^n = 0$, yet produces a finite response for every transfer. For GP2, for example, $\Delta C_L = +2.539 \times 10^{-3}$ in the older-only case, compared with -7.575×10^{-3} for the full-history case. A metric based only on the latest transferred state would therefore predict zero forcing for a case that demonstrably changes the next physical step.

The measured response ratios are consistent with the coefficients in $\eta^n = 2\epsilon^n - \epsilon^{n-1}/2$. Across the seven tested transfers, the latest/full state-norm ratio is 1.3330–1.3333 and the older/full magnitude is 0.3335–0.3341. State-vector superposition defects are 1.8–3.5%; lift superposition defects are 0.28–1.49%. The test therefore supports the directional linearization used in Sec. 3, but does not validate a general target resolvent for arbitrary perturbation directions.

The common-trial residual check also exposes an implementation detail that is hidden by a scalar norm. NN and BAR variants recover the predicted history forcing to approximately 10^{-7} relative error. For GP2 and GP3, the larger whole-vector discrepancy is localized to the 370 slip-wall vertices and to wall-normal momentum. The measured normal mismatch equals the projected normal forcing (2.125×10^{-6} for GP2 and 6.066×10^{-7} for GP3), while the tangential mismatch is only $O(10^{-11})$. The receiving solver removes wall-normal momentum through the slip condition; after this boundary admissibility operation, the tangential/interior identity is again satisfied at approximately 10^{-7} . Thus the algebraic BDF2 identity remains the correct starting point, but the physically admissible restart pipeline includes boundary-condition projection as well as interpolation.

Applying GC to the steady NACA transfer changes neither this conclusion nor the aerodynamic response materially. For NN1 the full-history lift response changes from 5.7873×10^{-3} to 5.7924×10^{-3} ; for BAR2 it changes from -7.2852×10^{-3} to -7.2850×10^{-3} . In these cases, enforcing the source integral alters a scalar component of the transferred field while leaving the force-relevant spatial discrepancy essentially intact.

5.5 Later aerodynamic response depends on the receiving flow problem

Two Euler experiments show that the first-step response is not the complete aerodynamic consequence of a transfer. Over the 400-step Euler window (approximately 0.047 chord flow-through times), GP2 has reproducible positive drag trends of 6.84×10^{-6} and 6.38×10^{-6} per 100 physical steps in the two realizations, whereas NN2 gives 3.95×10^{-5} and 3.84×10^{-5} . The two-seed mean ratio is about 5.8. This is a resolved difference for that Euler configuration, not a general ranking: lift trends are much less repeatable and the RANS calculations below do not reproduce the same method separation.

A single spatially localized GP2 transfer provides an independent test of delayed propagation. Increasing the launch distance from $0.05c$ to $0.20c$ delays the lift response, and pressure-shell measurements on the long $0.20c$ case give threshold-dependent propagation speeds between 1.09 and 1.27 for the 10–75% crossings after excluding the immediate transfer footprint. These values bracket the nondimensional acoustic speed $a_\infty = 1.183$. The spread is sensitivity to the chosen detector, not statistical uncertainty. The result supports a dominant acoustic-scale propagation in the Euler case and explains why a force response can peak after the transfer rather than at the first converged step. In particular, NN1 reaches a lift discrepancy 4.66 times its first-step value five steps after restart, whereas the second- and third-order cases have their largest recorded response at the first sampled step and then relax.

5.6 Fixed-body RANS: integral preservation improves, but a load ranking is not resolved

The RANS extension tests whether the preceding conclusions survive when turbulence histories and viscous stresses are part of the receiving problem. The runtime logs establish a non-dimensional physical step of $\Delta t^* = 0.347224$ for the configured 10^{-3} s step; with $U_\infty^* = 0.1$, 400 steps span 13.9 chord flow-through times. This is therefore not the short-time interval implied by treating the configured dimensional step as already nondimensional.

The transfer-integral comparison is unambiguous. Table 3 reports the maximum instantaneous source-to-target defect across the repeated-remeshing histories. GP2 reduces the monitored mean-flow integral defects by about two orders of magnitude relative to NN2. The same improvement occurs for the transported turbulence inventories: the maximum defect is reduced by a factor of 247 for $\rho\tilde{v}$ in SA and by factors of about 37 for both ρk and $\rho\omega$ in SST. These are transfer-event diagnostics; for SST the archived integral is evaluated before the primitive-variable positivity floor is applied. Turbulence production, destruction, and transport subsequently change the corresponding integrals and are not counted as remapping errors.

The aerodynamic comparison does not show an analogous method separation. The RMS total-drag deviations from each no-remeshing reference are $(6.07\text{--}6.79) \times 10^{-5}$ across the eight method/seed/closure combinations. Paired NN2–GP2 differences have RMS $(0.77\text{--}1.11) \times 10^{-5}$, and their ordering changes with closure and random realization. On these raw statistics alone, NN2 and GP2 therefore have similar aggregate drag errors despite markedly different integral defects.

Three numerical controls on steps 2–40 of the seed-1 sequence (three transfer events, identical displacement files verified by hash) bound what this comparison can resolve (Table 8). The dual-time configuration allows up to 800 inner iterations and targets $\log_{10}(\text{rms } \rho) = -12$; in the repeated-remeshing runs 144–171 of 398 SA steps and 277–333 of 398 SST steps stop at that cap, with median terminal SST density residuals of $10^{-11.1}\text{--}10^{-11.3}$. Doubling the cap to 1600 brings every SA step to the target and leaves 28 of 38 SST steps at the cap; it changes the case-minus-reference lift and drag by 0.2–0.3% of their RMS for SA and by 1–4% for SST, and the paired NN2–GP2

Table 3: Maximum absolute source-relative integral defect during the repeated-remeshing RANS runs. Values compare the source and returned target states at a transfer event; the final column is the ratio of the NN2 and GP2 maxima.

Closure	Inventory	NN2	GP2	NN2/GP2
SA	rho	$8.41 \cdot 10^{-11}$	$4.57 \cdot 10^{-13}$	184
SA	rho u	$3.78 \cdot 10^{-9}$	$5.76 \cdot 10^{-11}$	66
SA	rho v	$1.65 \cdot 10^{-8}$	$8.46 \cdot 10^{-11}$	195
SA	rho E	$9.54 \cdot 10^{-11}$	$5.83 \cdot 10^{-13}$	164
SA	rho nu-tilde	$1.7 \cdot 10^{-6}$	$6.88 \cdot 10^{-9}$	247
SST	rho	$8.25 \cdot 10^{-11}$	$4.68 \cdot 10^{-13}$	177
SST	rho u	$3.65 \cdot 10^{-9}$	$2.89 \cdot 10^{-11}$	126
SST	rho v	$1.64 \cdot 10^{-8}$	$7.78 \cdot 10^{-11}$	211
SST	rho E	$9.37 \cdot 10^{-11}$	$6.75 \cdot 10^{-13}$	139
SST	rho k	$2.22 \cdot 10^{-6}$	$6 \cdot 10^{-8}$	37
SST	rho omega	$7.98 \cdot 10^{-8}$	$2.15 \cdot 10^{-9}$	37

difference by 8–14%. The production loads are therefore converged with respect to the inner loop well below the paired difference. Halving the physical step with the same three events replayed at the same physical times leaves the RMS of the SA case-minus-reference response within 5%, but the response changes step by step by 25–40% of its RMS for both transfers alike, and the paired NN2–GP2 difference changes by about 140% of its own RMS while keeping its magnitude. The paired difference is thus below the temporal resolution of the remeshing transient. Finally, a finer viscous mesh with the same first off-wall spacing changes the no-transfer lift by -0.8% (SA) and -1.0% (SST) and the drag by -1.8% and -1.6% : the spatial discretization uncertainty of the absolute loads, $1.5\text{--}1.8 \times 10^{-4}$ in drag, exceeds the remeshing response itself, so no absolute aerodynamic accuracy is claimed. The RANS calculations therefore establish that the two transfer pipelines produce different returned turbulence states and substantially different integral defects, but they do not resolve a quantitative aerodynamic ranking between NN2 and GP2.

This limitation is itself informative for the paper’s central question. Transfer accuracy cannot be assessed independently of the receiving solve: once the interpolation difference falls below the temporal sensitivity of the physical step, a nominally more conservative transfer need not produce a measurable load improvement. The RANS results should therefore be read as an application-level consistency test and as a bound on what can presently be distinguished, rather than as evidence that NN2 and GP2 are aerodynamically equivalent.

6 Discussion and conclusions

The recomputed results support a narrower and stronger conclusion than a universal ranking of interpolation schemes. Remeshing introduces several distinct errors: approximation of the source field on the new mesh, violation of bounds or admissibility, changes in monitored integrals, incompatibility of the multistep histories with the receiving discretization, and the subsequent response of the flow solver. No single scalar diagnostic orders all of these effects.

First, global conservation is demonstrably insufficient as a spatial-quality criterion. In the Riemann stress test, increasing remeshing frequency or deformation amplitude primarily broadens the density transition. NN1 reaches a width increase of 21–26 cells and BAR2 3–23 cells, while their position errors remain sub-cell in the recomputed extraction. Correcting NN1 reduces the cumulative density-integral defect to roundoff but leaves 18–26 cells of excess width. A global

integral constraint therefore fixes the total amount of a conserved quantity without determining its distribution. Higher-order NN/BAR variants and Galerkin projection remain much less diffusive in the same experiment, so the evidence does not justify attributing the result uniquely to “local conservation.”

Second, a BDF2 restart must be assessed as a two-history problem. The older-only NACA configuration produces a finite state and force response even though the latest history is exact, and the latest/full and older/full response ratios recover the BDF2 coefficients to within a few tenths of a percent in norm. The remaining 1.8–3.5% vector superposition defects delimit the accuracy of the local linear interpretation. The residual audit further shows that boundary-condition enforcement is part of the transfer pipeline: Galerkin projection creates a wall-normal momentum history component that the slip-wall condition removes. A transfer method should therefore be evaluated after all boundary, limiting, positivity, and representation operations required by the receiving solver, not only through the ideal interpolation operator.

Third, the aerodynamic response is problem dependent. In the repeated Euler airfoil calculation GP2 reduces the reproducible drag trend by approximately a factor of 5.8 relative to NN2, and the localized-transfer experiment exhibits an acoustic-scale delayed response. The fixed-body RANS calculations do not reproduce an equally clear method separation. GP2 reduces the monitored turbulence-inventory defects by factors of roughly 37–247, yet the NN2 and GP2 drag histories have similar aggregate RMS deviations. The paired drag and lift differences are converged with respect to the dual-time iteration cap but change by more than their own magnitude when the physical step is halved, and the absolute loads carry a spatial-discretization uncertainty larger than the remeshing response; they are therefore not resolved as an interpolation-method ranking.

The practical implication is that transfer order and integral conservation should be treated as necessary diagnostics, not as stand-alone selection criteria. A robust remeshing workflow should verify smooth-field accuracy, boundedness and admissibility, source-to-target integral changes, compatibility of every multistep history, boundary processing, and the convergence of the receiving physical step. When the aerodynamic difference between transfer methods is smaller than the temporal and spatial discretization sensitivity of the receiving solve, improving the interpolation alone cannot be shown to improve the final load prediction.

Within these limits, the paper establishes four results: the implemented transfers recover their intended smooth-field orders; global correction can enforce a solver integral without controlling repeated-remapping diffusion; both BDF2 history levels contribute directly to the restart transient; and improved integral preservation does not imply a universal improvement in aerodynamic loads. The Euler evidence for these mechanisms is internally consistent and the recomputed Riemann profile diagnostics remove the earlier sampling-dependent position-drift narrative. The RANS extension demonstrates applicability to transported turbulence variables and viscous forces; its numerical controls show that the aerodynamic method comparison is limited by the temporal resolution of the remeshing transient and by the spatial discretization of the baseline, not by the inner iteration, so the defensible RANS claim is limited to transfer-state and integral behavior rather than comparative load accuracy.

A Supporting recomputed results

A.1 Vertex restriction of a quadratic projection

The conservation property of the complete Galerkin field does not automatically transfer to a vertex-only restart representation. Consider $f(x) = x^2$ on a uniform partition of $[0, 1]$. A quadratic finite-element space represents f exactly, so the P^2 projection itself has no approximation error.

If only the vertex values are retained and integrated with the corresponding P^1 nodal weights, however,

$$Q_h(f) = \frac{1}{3} + \frac{h^2}{6}, \quad (30)$$

while the exact integral is $1/3$. Thus an exact quadratic projected field can acquire a second-order lumped-integral discrepancy solely through representation change. This is why the complete Galerkin field, the vertex restart state, and any subsequent correction are treated as distinct objects in the main text.

Riemann frequency and amplitude matrix

Table 4 gives the complete recomputed frequency comparison used to support Sec. 5.3. Position and width use one common extraction procedure within this dataset.

Table 4: Recomputed transition position and width changes for the 80- and 400-transfer Riemann runs at SF4.

Transfer	Transfers	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho$
NN1	80	0.14	0.57	$1.1 \cdot 10^{-4}$
NN1	400	0.1	21.41	$5 \cdot 10^{-4}$
NN2	80	-0.02	-0.01	$-8.8 \cdot 10^{-6}$
NN2	400	-0.04	0	$-1.7 \cdot 10^{-5}$
NN3	80	0.01	-0.01	$2.2 \cdot 10^{-5}$
NN3	400	0.02	0	$2.7 \cdot 10^{-5}$
BAR2	80	0.01	0.25	$2 \cdot 10^{-5}$
BAR2	400	-0.03	3.24	$1.4 \cdot 10^{-4}$
BAR3	80	0	0.01	$-1.7 \cdot 10^{-5}$
BAR3	400	0	0.07	$-5.2 \cdot 10^{-6}$
GP2	80	0.01	0	$2.6 \cdot 10^{-6}$
GP2	400	0.01	0	$-3 \cdot 10^{-6}$
GP3	80	-0.01	0	$-2.1 \cdot 10^{-5}$
GP3	400	-0.01	0.02	$1.8 \cdot 10^{-5}$

NACA history-response ratios

Table 5: Directional BDF2 response ratios and superposition defects. The ideal linear-history values are $4/3$ for latest/full and $1/3$ in magnitude for older/full.

Transfer	e_{sup}^U	$\ \delta_n\ /\ \delta_f\ $	$\ \delta_{n-1}\ /\ \delta_f\ $	$e_{sup}^{C_L}$	$\Delta C_{L,n}/\Delta C_{L,f}$	$\Delta C_{L,n-1}/\Delta C_{L,f}$
NN1	$1.81 \cdot 10^{-2}$	1.3333	0.3335	$1.49 \cdot 10^{-2}$	1.3329	-0.3179
NN2	$3.53 \cdot 10^{-2}$	1.3331	0.3341	$4.2 \cdot 10^{-3}$	1.3318	-0.336
NN3	$3.4 \cdot 10^{-2}$	1.3331	0.334	$5.29 \cdot 10^{-3}$	1.3309	-0.3362
BAR2	$3.36 \cdot 10^{-2}$	1.3332	0.334	$3.05 \cdot 10^{-3}$	1.3328	-0.3358
BAR3	$3.34 \cdot 10^{-2}$	1.3332	0.334	$2.83 \cdot 10^{-3}$	1.3329	-0.3357
GP2	$3.39 \cdot 10^{-2}$	1.333	0.334	$2.98 \cdot 10^{-3}$	1.3322	-0.3352
GP3	$3.42 \cdot 10^{-2}$	1.3331	0.334	$3.26 \cdot 10^{-3}$	1.3326	-0.3359

Riemann deformation realizations

Table 6 repeats the 400-transfer stress comparison of Sec. 5.3 with two further random deformation sequences (seeds 2 and 3) at both amplitudes.

Table 6: Transition position error and width change (cells) after 400 transfers for three independent deformation realizations; SF2 permits twice the displacement of SF4.

Transfer	SF	Seed	$\Delta x/h$	$\Delta w/h$
BAR2	2	1	-0.06	22.91
BAR2	2	2	-0.03	22.98
BAR2	2	3	-0.07	22.9
BAR2	4	1	-0.03	3.24
BAR2	4	2	0	3.2
BAR2	4	3	-0.04	3.12
GP2	2	1	0.17	0.06
GP2	2	2	0.19	0.06
GP2	2	3	0.19	0.03
GP2	4	1	0.01	0
GP2	4	2	0.02	0
GP2	4	3	0.01	0.01
NN1	2	1	-0.38	26.36
NN1	2	2	-0.29	24.98
NN1	2	3	0.44	27.14
NN1	4	1	0.1	21.41
NN1	4	2	0.3	22.89
NN1	4	3	0.2	22.75

RANS numerical-control diagnostics

The runtime logs give $\Delta t^* = 0.347224$ and $U_\infty^* = 0.1$. Table 7 reports how often the repeated-remeshing calculations reach the 800-iteration dual-time cap. These counts motivate the conservative interpretation of the load comparison in Sec. 5.6.

Table 7: Inner-iteration convergence of the repeated-remeshing RANS calculations. “At cap” counts physical steps that reach the maximum inner-iteration count before satisfying the configured density-residual criterion.

Closure	Transfer	Seed	Steps	At cap	median $\log_{10} \text{rms}(\rho)$
SA	GP2	1	398	144	-12
SA	GP2	2	398	169	-12
SA	NN2	1	398	156	-12
SA	NN2	2	398	171	-12
SST	GP2	1	398	333	-11.1
SST	GP2	2	398	284	-11.31
SST	NN2	1	398	323	-11.24
SST	NN2	2	398	277	-11.24

Table 8 gives the two solver controls on steps 2–40 of the seed-1 sequence: the change of the case-minus-reference response D_m and of the paired difference $P = \text{NN2} - \text{GP2}$ between the production setting and the control, as RMS over the window and relative to the production RMS. Table 9 gives the no-transfer steady loads on the production and the finer viscous mesh.

Table 8: Sensitivity of the repeated-remeshing RANS response to the dual-time iteration cap (800 to 1600) and to the physical step (halved, events replayed at the same physical times; SA only). D_m is the case-minus-reference response of transfer m , P the paired NN2–GP2 difference.

Closure	Control	Quantity	Coefficient	RMS, production	RMS of change	Change / production
SA	cap1600	GP2	CL	$2.63 \cdot 10^{-4}$	$6.53 \cdot 10^{-7}$	0.002
SA	cap1600	GP2	CD	$6.56 \cdot 10^{-5}$	$1.33 \cdot 10^{-7}$	0.002
SA	cap1600	GP2	CD _v	$2.13 \cdot 10^{-7}$	$7.69 \cdot 10^{-9}$	0.036
SA	cap1600	NN2	CL	$2.6 \cdot 10^{-4}$	$8.79 \cdot 10^{-7}$	0.003
SA	cap1600	NN2	CD	$6.61 \cdot 10^{-5}$	$6.69 \cdot 10^{-8}$	0.001
SA	cap1600	NN2	CD _v	$2.2 \cdot 10^{-7}$	$5.29 \cdot 10^{-9}$	0.024
SA	cap1600	P	CL	$1.21 \cdot 10^{-5}$	$1.32 \cdot 10^{-6}$	0.109
SA	cap1600	P	CD	$2.19 \cdot 10^{-6}$	$1.83 \cdot 10^{-7}$	0.084
SA	cap1600	P	CD _v	$2.97 \cdot 10^{-8}$	$1.1 \cdot 10^{-8}$	0.371
SA	halfstep	GP2	CL	$2.63 \cdot 10^{-4}$	$1.11 \cdot 10^{-4}$	0.423
SA	halfstep	GP2	CD	$6.56 \cdot 10^{-5}$	$1.6 \cdot 10^{-5}$	0.244
SA	halfstep	GP2	CD _v	$2.13 \cdot 10^{-7}$	$8.12 \cdot 10^{-7}$	3.817
SA	halfstep	NN2	CL	$2.6 \cdot 10^{-4}$	$1.06 \cdot 10^{-4}$	0.408
SA	halfstep	NN2	CD	$6.61 \cdot 10^{-5}$	$1.58 \cdot 10^{-5}$	0.24
SA	halfstep	NN2	CD _v	$2.2 \cdot 10^{-7}$	$8.11 \cdot 10^{-7}$	3.685
SA	halfstep	P	CL	$1.21 \cdot 10^{-5}$	$1.67 \cdot 10^{-5}$	1.386
SA	halfstep	P	CD	$2.19 \cdot 10^{-6}$	$3.21 \cdot 10^{-6}$	1.468
SA	halfstep	P	CD _v	$2.97 \cdot 10^{-8}$	$3.95 \cdot 10^{-8}$	1.332
SST	cap1600	GP2	CL	$1.65 \cdot 10^{-4}$	$6.17 \cdot 10^{-6}$	0.037
SST	cap1600	GP2	CD	$5.45 \cdot 10^{-5}$	$6.15 \cdot 10^{-7}$	0.011
SST	cap1600	GP2	CD _v	$8.11 \cdot 10^{-7}$	$2.04 \cdot 10^{-7}$	0.251
SST	cap1600	NN2	CL	$1.65 \cdot 10^{-4}$	$6 \cdot 10^{-6}$	0.036
SST	cap1600	NN2	CD	$5.51 \cdot 10^{-5}$	$5.73 \cdot 10^{-7}$	0.01
SST	cap1600	NN2	CD _v	$8.02 \cdot 10^{-7}$	$2.07 \cdot 10^{-7}$	0.258
SST	cap1600	P	CL	$5.98 \cdot 10^{-6}$	$8.19 \cdot 10^{-7}$	0.137
SST	cap1600	P	CD	$1.59 \cdot 10^{-6}$	$9.59 \cdot 10^{-8}$	0.06
SST	cap1600	P	CD _v	$1.5 \cdot 10^{-8}$	$9.46 \cdot 10^{-9}$	0.633

Table 9: No-transfer steady loads on the production mesh (23,271 vertices) and on a finer viscous mesh with the same first off-wall spacing (32,102 vertices).

Closure	Coefficient	Production mesh	Finer mesh	Difference	%
SA	CL	0.676257	0.670838	$-5.42 \cdot 10^{-3}$	-0.8
SA	CD	0.010229	0.010048	$-1.81 \cdot 10^{-4}$	-1.77
SA	CD _p	0.003559	0.003401	$-1.58 \cdot 10^{-4}$	-4.44
SA	CD _v	0.006671	0.006647	$-2.4 \cdot 10^{-5}$	-0.36
SA	CM _z	0.000671	-0.00053	$-1.2 \cdot 10^{-3}$	-179.07
SST	CL	0.675605	0.668596	$-7.01 \cdot 10^{-3}$	-1.04
SST	CD	0.009858	0.009705	$-1.53 \cdot 10^{-4}$	-1.55
SST	CD _p	0.003469	0.003321	$-1.48 \cdot 10^{-4}$	-4.27
SST	CD _v	0.006389	0.006385	$-4 \cdot 10^{-6}$	-0.06
SST	CM _z	0.000382	-0.001155	$-1.54 \cdot 10^{-3}$	-402.19

B Data provenance

The interpolation-order plots retain the existing generated data. All other numerical statements in the rewritten Results and Conclusions were re-assessed against the recalculated CSVs in the repository rather than inherited from the previous manuscript narrative. Existing source CSV paths and filenames are unchanged. The additional `reassessed_*` files in `generated/` are compact views used only for manuscript tables; they do not replace their source data. The RANS runtime and convergence statements use the executed-run startup and inner-iteration records; the RANS numerical controls and the finer-mesh baseline are separate executed runs with their own force histories and surface exports; the Riemann realizations are separate 400-transfer runs evaluated with the same extraction as the seed-1 matrix. The Euler NACA boundary-condition interpretation uses the double-precision residual-probe and wall-split diagnostics. No CFD result was inferred from the previous prose when a recomputed data product was available.

References

- [1] Frédéric Alauzet, Adrien Loseille, and Géraldine Olivier. “Time-Accurate Multi-Scale Anisotropic Mesh Adaptation for Unsteady Flows in CFD”. In: *Journal of Computational Physics* 373 (2018), pp. 28–63. DOI: [10.1016/j.jcp.2018.06.043](https://doi.org/10.1016/j.jcp.2018.06.043).
- [2] Bastien Sauvage, Frédéric Alauzet, and Alain Dervieux. “A space and time fixed point mesh adaptation method”. In: *Journal of Computational Physics* 519 (2024), p. 113389. DOI: [10.1016/j.jcp.2024.113389](https://doi.org/10.1016/j.jcp.2024.113389).
- [3] Patrick E. Farrell and James R. Maddison. “Conservative Interpolation between Volume Meshes by Local Galerkin Projection”. In: *Computer Methods in Applied Mechanics and Engineering* 200.1–4 (2011), pp. 89–100. DOI: [10.1016/j.cma.2010.07.015](https://doi.org/10.1016/j.cma.2010.07.015).
- [4] Xiangmin Jiao and Michael T. Heath. “Common-Refinement-Based Data Transfer between Non-Matching Meshes in Multiphysics Simulations”. In: *International Journal for Numerical Methods in Engineering* 61 (2004), pp. 2402–2427.
- [5] J. R. Maddison and H. R. Hiester. “Optimal Constrained Interpolation in Mesh-Adaptive Finite Element Modeling”. In: *SIAM Journal on Scientific Computing* 39.5 (2017), A2257–A2286. DOI: [10.1137/15M102054X](https://doi.org/10.1137/15M102054X).
- [6] J. R. Maddison, C. J. Cotter, and P. E. Farrell. “Geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling”. In: *Ocean Modelling* 37.1–2 (2011), pp. 35–48. DOI: [10.1016/j.ocemod.2010.12.007](https://doi.org/10.1016/j.ocemod.2010.12.007).
- [7] J. R. Maddison, C. J. Cotter, and P. E. Farrell. “Corrigendum to “Geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling” [Ocean Model. 37 (2011) 35–48]”. In: *Ocean Modelling* 68 (2013), p. 106. DOI: [10.1016/j.ocemod.2013.04.007](https://doi.org/10.1016/j.ocemod.2013.04.007).
- [8] Timothy J. Barth and Paul O. Frederickson. “Higher Order Solution of the Euler Equations on Unstructured Grids Using Quadratic Reconstruction”. In: *28th Aerospace Sciences Meeting*. AIAA Paper 90-0013. Reno, Nevada, 1990.
- [9] Timothy J. Barth. “Recent Developments in High Order k -Exact Reconstruction on Unstructured Meshes”. In: *31st Aerospace Sciences Meeting*. AIAA Paper 93-0668. Reno, Nevada, 1993.

- [10] Dimitri J. Mavriplis. “Revisiting the Least-Squares Procedure for Gradient Reconstruction on Unstructured Meshes”. In: *16th AIAA Computational Fluid Dynamics Conference*. AIAA Paper 2003-3986. Orlando, Florida, 2003. DOI: [10.2514/6.2003-3986](https://doi.org/10.2514/6.2003-3986).
- [11] Boris Diskin and James L. Thomas. *Accuracy of Gradient Reconstruction on Grids with High Aspect Ratio*. Tech. rep. NIA Report 2008-12. Hampton, Virginia: National Institute of Aerospace, 2008.
- [12] Timothy J. Barth and Dennis C. Jespersen. “The Design and Application of Upwind Schemes on Unstructured Meshes”. In: *27th Aerospace Sciences Meeting*. AIAA Paper 89-0366. Reno, Nevada, 1989. DOI: [10.2514/6.1989-366](https://doi.org/10.2514/6.1989-366).
- [13] V. Venkatakrishnan. “Convergence to Steady State Solutions of the Euler Equations on Unstructured Grids with Limiters”. In: *Journal of Computational Physics* 118.1 (1995), pp. 120–130.
- [14] Kristopher Michalak and Carl Ollivier-Gooch. “Accuracy Preserving Limiter for the High-Order Accurate Solution of the Euler Equations”. In: *Journal of Computational Physics* 228.23 (2009), pp. 8693–8711.
- [15] Hiroaki Nishikawa. “New Unstructured-Grid Limiter Functions”. In: *AIAA SciTech Forum*. AIAA Paper 2022-1374. 2022. DOI: [10.2514/6.2022-1374](https://doi.org/10.2514/6.2022-1374).
- [16] David A. Dunavant. “High Degree Efficient Symmetrical Gaussian Quadrature Rules for the Triangle”. In: *International Journal for Numerical Methods in Engineering* 21.6 (1985), pp. 1129–1148.
- [17] Patrick Keast. “Moderate-Degree Tetrahedral Quadrature Formulas”. In: *Computer Methods in Applied Mechanics and Engineering* 55 (1986), pp. 339–348.
- [18] Mikhail Shashkov and Burton Wendroff. “The Repair Paradigm and Application to Conservation Laws”. In: *Journal of Computational Physics* 198.1 (2004), pp. 265–277.
- [19] Milan Kucharik, Mikhail Shashkov, and Burton Wendroff. “An Efficient Linearity-and-Bound-Preserving Remapping Method”. In: *Journal of Computational Physics* 188.2 (2003), pp. 462–471.
- [20] Bruno Koobus and Charbel Farhat. “Second-Order Time-Accurate and Geometrically Conservative Implicit Schemes for Flow Computations on Unstructured Dynamic Meshes”. In: *Computer Methods in Applied Mechanics and Engineering* 170.1–2 (1999), pp. 103–129. DOI: [10.1016/S0045-7825\(98\)00207-2](https://doi.org/10.1016/S0045-7825(98)00207-2).
- [21] Philippe Geuzaine, Céline Grandmont, and Charbel Farhat. “Design and Analysis of ALE Schemes with Provable Second-Order Time-Accuracy for Inviscid and Viscous Flow Simulations”. In: *Journal of Computational Physics* 191.1 (2003), pp. 206–227. DOI: [10.1016/S0021-9991\(03\)00311-5](https://doi.org/10.1016/S0021-9991(03)00311-5).
- [22] Michael B. Giles and Niles A. Pierce. “An Introduction to the Adjoint Approach to Design”. In: *Flow, Turbulence and Combustion* 65 (2000), pp. 393–415. DOI: [10.1023/A:1011430410075](https://doi.org/10.1023/A:1011430410075).
- [23] Philippe R. Spalart and Steven R. Allmaras. “A One-Equation Turbulence Model for Aerodynamic Flows”. In: *La Recherche Aéronautique* 1 (1994), pp. 5–21.
- [24] Florian R. Menter. “Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications”. In: *AIAA Journal* 32.8 (1994), pp. 1598–1605. DOI: [10.2514/3.12149](https://doi.org/10.2514/3.12149).